

Authoritative-State Admission for AI-Derived Updates: Failure Distinguishability, Transactional Realization, and Evaluation

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Abstract

AI-derived record updates must distinguish a machine proposal from the value a human authorizes. We specify *authoritative-state admission*: authorization binds an exact persisted candidate to an explicit value, Correction preserves the original proposal, and the complete successor retains every field’s source. Five safe/unsafe history pairs establish class-wise irredundancy relative to a declared observation model. The Python/SQLite realization combines candidate-bound authorization, compare-and-swap-protected admission, and reverse-trace validation, checked through bounded Alloy analysis, formal–concrete projection, and a 35-case fault catalogue. The evaluated transactional value-audit safeguards do not by themselves entail an explicit authorization-to-exact-candidate relation. Both the frozen value-audit control and a strengthened transactional value-audit control agree with candidate-bound admission on seven of eight single-history cases; equal-valued candidate substitution separates them. Richer recorded review context narrows this distinction, while exact binding discriminates candidates equivalent under those observations. Frozen v8 workloads characterize admission and trace costs. Across three qualified model configurations, 320/360 planned benign runs met the strict-trajectory criterion; 25 lacked a scorable verdict. Fixed host checks supply separate integration-conformance observations, with all runtime failures retained. These results concern the declared model and evaluated system; host authentication and review-channel integrity remain external assumptions.

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1. Introduction

AI-generated updates can become persistent facts used for reporting, payment, scheduling, or later decisions. Admission must therefore establish which authority approved which value from which persisted proposal and record state.

Consider a record whose authoritative quantity is 90. A recognizer proposes 100, but a reviewer authorizes 101; an unchanged batch-code field should retain its earlier source. Accepting a machine proposal of 101 would produce the same final value with different attribution. Rewriting the candidate would falsely attribute the correction to the model, while dropping the unchanged field’s source would preserve values but lose lineage. Admission must retain the predecessor 90, the proposal $x_c = 100$, and the authorized value $x_a = 101$, allowing

$$x_c \neq x_a \tag{1}$$

with both value roles and the complete successor’s sources intact.

The target systems permit human correction of AI-generated field candidates, persist accepted updates as operational facts, and require later reconstruction of authorization and origin. Existing activation, governed-memory, and transaction mechanisms provide adjacent foundations: Continuity Kernel separates candidates from atomic activation; LatticeMind handles claim selection and supersession; authority-collapse work studies lost source constraints; MutMem binds authorized changes to predecessor history; and MemTxn supports source-grounded updates and recovery (He and Yu, 2026a; Zhou et al., 2026; Zhan et al., 2026; Saidi, 2026; Cui et al., 2026). We specialize this problem to the following correction-aware, field-level relation:

$$\begin{aligned} \textit{exact candidate} &\rightarrow \textit{candidate-bound human authorization} \\ &\rightarrow \textit{explicit authorized value} \rightarrow \textit{complete successor} \tag{2} \\ &\rightarrow \textit{total field-source attribution.} \end{aligned}$$

The candidate remains historical evidence; human authorization supplies value authority; a trusted transaction constructs the successor. Ordinary transaction and logging facilities can implement this relation. The engineering

increment is their joint, checkable binding of proposal identity, a possibly different authorized value, and exact changed/unchanged field sources.

We ask which observable distinctions preserve that relation against candidate substitution, correction erasure, stale replay, fragmented successors, and source ambiguity. Under a declared failure and observation model, paired histories expose what becomes indistinguishable when each information class is omitted. We study the relation through the Python/SQLite reference realization AUTO-DECTE:

- C1: Correction-aware admission contract.** Bind authorization to the exact candidate, keep proposed and authorized values separately attributable, and retain total field sources in one complete successor.
- C2: Conditional failure distinguishability.** Five safe/unsafe history pairs establish class-wise irredundancy within the declared failure and observation model.
- C3: Relational and transactional realization.** Connect the contract to bounded Alloy checks, a CAS-protected transaction, selected formal-concrete projections, and an executable fault catalogue.
- C4: Cost and integration evidence.** Characterize frozen v8 persistence-equivalent admission and trace costs, and separate restricted-agent task outcomes from host-enforced authority checks across three qualified configurations.
- C5: Evaluated-control separation.** Equal-valued candidate substitution separates candidate-bound admission from both evaluated transactional value-audit controls, which agree with it on the other seven single-history cases. The separation persists with strengthened safeguards and narrows with richer recorded review context.

Constructed workflow example. An operations team needs to identify which persisted proposal a reviewer inspected. Two extraction retries may have the same record, field, and authorized value but different timestamps and evidence locators. Value-level approval suffices when the audit question is whether a value was approved. Instance-level accountability additionally asks which artifact was reviewed; candidate-bound admission records that instance and rejects substitution. The need to distinguish retries depends on

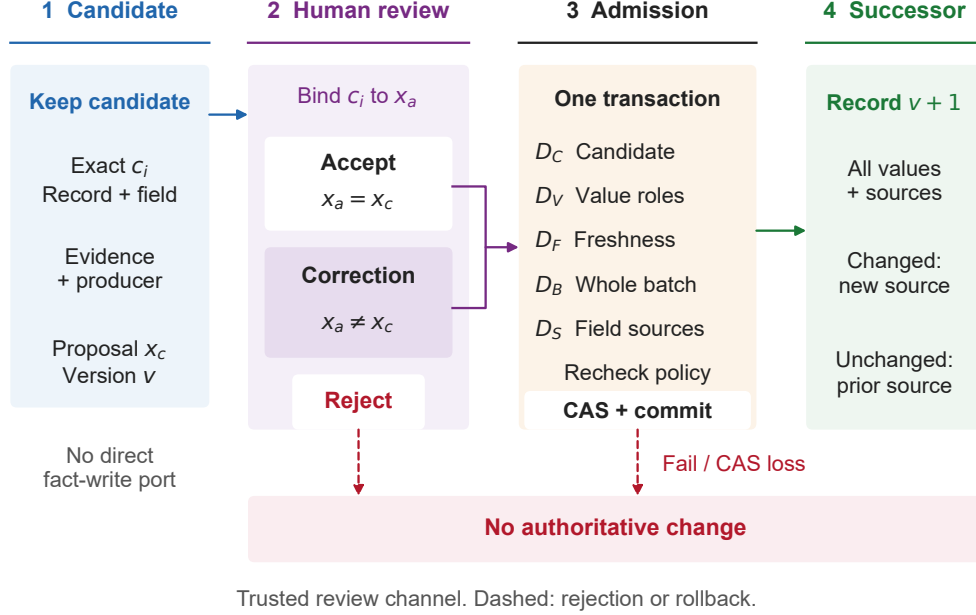


Figure 1: Correction-aware authoritative-state admission. Only candidate-bound Accept or Correction decisions proceed to transactional admission. The declared distinction checks, host-policy revalidation, and version compare-and-swap protect creation of a complete successor and total source map. Reject, failed validation, or a lost compare-and-swap leaves authoritative state unchanged. Authorization records responsibility, not factual truth. AI-assisted diagram scripting: OpenAI Codex (gpt-6-astra, OpenAI, September 2026 revision); deterministic vector rendering of the declared workflow.

the accountability policy. The current path records value-changing updates; renewed review without a value change needs a separate review-event path.

The contract records and rechecks authority under externally supplied host authentication and review-channel integrity; the trust boundary is detailed in section 3.

The paper proceeds from the admission relation (section 3) to relational encodings (RQ1–RQ2), persisted behavior and controls (RQ3–RQ5), and cost and integration (RQ6–RQ8). Figure 1 shows the admission boundary.

2. Background and related work

Authoritative-state admission sits at the intersection of agent-state governance, authorization, transactional provenance, and formal-concrete checking.

We organize prior work by the software boundary it governs: memory activation and update, authorization and effects, provenance, or executable validation. The comparison includes public preprints available through 3 September 2026, with publication status recorded separately.

2.1. Authoritative state and governed memory

Continuity Kernel is the closest general activation-boundary neighbor. It separates off-commit candidate evaluation from a short transaction that revalidates ownership, predecessor authority, freshness, and effect uniqueness before atomically installing a complete accepted unit (He and Yu, 2026a).

LatticeMind moves claim conflict handling into write-time structured memory. It represents proposed, confirmed, contested, and superseded items; reconciles slot-scoped claims using symbolic checks and selective model assistance; and retains losing claims with provenance (Zhou et al., 2026). Its reviewed artifact focuses on conflict-aware claim selection and supersession.

When Memory Becomes Authority identifies *authority collapse*: consolidation may preserve claim content while erasing source constraints governing later reuse (Zhan et al., 2026). Correct Is Not Governed similarly separates correct outcomes from governed execution through versioned authority/fact sets and dependency provenance (Salas, 2026). These works establish that retaining content or reaching the right answer does not by itself preserve authority.

MutMem cryptographically authorizes mutation of an already persistent memory property, binding old/new retrieval weights, signer epoch, provenance, and a no-fork predecessor (Saidi, 2026). MemTX stages and validates belief writes, while the distinct MemTxn system places source-supported updates, conflict-conditioned visibility, and complete-state recovery behind an answer-model-external transaction boundary (Li et al., 2026; Cui et al., 2026). SuperLocalMemory 4.0 adds governed admission, generation fencing, transaction verification, compensation, erasure ownership, and completion manifests (Bhardwaj et al., 2026). Together, these systems establish authorized mutation, update transactions, governed admission, source validation, and recovery as adjacent foundations.

Stored Is Not Supported separates persisted availability from warranted assertion through typed provenance, authorized projections, and a mediated release gate (He and Yu, 2026b). It takes an authenticated accepted-state head as input and treats lower-level atomic storage as an assumption. The

boundaries are complementary: valid admission alone does not establish that a later statement is supported or authorized for disclosure.

Table 1 compares reported admission relations. Atomicity and freshness alone do not bind a human-authorized value to its machine proposal or preserve unchanged-field sources. An unreported relation is not a demonstrated defect or an inability to extend the compared system.

When Stale Constraints Go Unchecked shows empirically that immutable provenance can remain reachable while an agent fails to revisit a superseded constraint under a verification budget (Nakayashiki, 2026). It motivates the freshness distinction used in our declared admission model, but does not itself define a candidate-bound authoritative commit relation.

2.2. Proof-carrying and transaction-bound authority

Proof-carrying authentication places a machine-checkable proof on the requester and checks it at the server with a minimal trusted computing base (Appel and Felten, 1999). DPoP sender-constrains OAuth tokens to a proof-of-possession key and the presenting HTTP request (Fett et al., 2023). The active OAuth Transaction Tokens draft propagates signed authorization and request context through one call chain, whereas the active individual SPT-Txn draft binds a short-lived token to a declared action and resource (Tulshibagwale et al., 2026; Coetzee, 2026). Both Internet-Drafts are works in progress. CapLease additionally treats authorization consumption as durable state for replay-resistant agent actions (Xu et al., 2026).

PCE separates proposed traces, certification, and execution; DTF derives execution authority from a structured intent, justification proof, and approval record; EBTE checks typed action claims against server-held intent, policy, payload, tool, risk, provenance, and freshness facts; and CAP+PCL composes provenance-bound context attestation with deterministic policy authorization (Yanglet et al., 2026; He and Yu, 2026c; Zhu and Wang, 2026; Chitan, 2026). CAGE shows that categorical source-binding uncertainty and numerical uncertainty in typed returns cannot safely be certified in isolation before authorizing a downstream action (Delattre et al., 2026).

ToolGate checks preconditions and postconditions before symbolic-state updates; CapChain governs field/value updates through scoped capabilities and the current provenance predecessor root (Liu et al., 2026; Choong et al., 2026).

Agent harnesses provide the extension boundary through which a model can reach such services. A source-level comparison of deepagents, pi, and

DeepSeek Harness reports explicit extension seams as a convergent feature (Dai, 2026); the pinned DeepSeek Harness architecture specifically registers model-facing capabilities on `ctx.tools` and routes execution through a guarded runtime pipeline (DeepSeek AI, 2026).

Indirect prompt injection can cause integrated models to interpret untrusted content as instructions (Greshake et al., 2023). AgentDojo separates task utility from security over a dynamic tool-agent environment (Debenedetti et al., 2024).

2.3. Versioned transaction provenance

Why- and where-provenance distinguish why an output exists from where its values originated, and later surveys systematize why-, how-, and where-provenance (Buneman et al., 2001; Cheney et al., 2009). The Open Provenance Model and W3C PROV provide technology-independent vocabularies for causal histories, entities, activities, agents, derivations, and responsibility (Moreau et al., 2011; Moreau and Missier, 2013). Transaction reenactment and multi-version provenance reconstruct how database states arise from update histories (Arab et al., 2016, 2018; Arab, 2019).

These provenance vocabularies can represent the relevant relationships; adopting a vocabulary alone does not specify when admission must enforce candidate binding, authorized-value equality, or exact source copy-forward.

2.4. Bounded analysis and executable projection

Alloy searches finite relational scopes through SAT-based model finding (Jackson, 2002; Torlak and Jackson, 2007). TestEra maps Alloy-generated inputs to Java executions and abstracts their outputs back to Alloy; bounded program checkers similarly translate annotated programs into finite relational or propositional searches (Marinov and Khurshid, 2001; Dennis et al., 2006; Galeotti et al., 2013).

Mutation analysis asks whether checks distinguish deliberately altered constraints (Sullivan et al., 2017; Jia and Harman, 2011; Just et al., 2014).

2.5. Stateful, concurrent, and reproducible evidence

Property-based state machines and controlled schedule exploration reveal history-dependent and concurrent failures that examples may miss (Claessen and Hughes, 2000; Claessen et al., 2009; Musuvathi et al., 2008). Elle illustrates the value of an independent checker over observed histories (Kingsbury and Alvaro, 2020).

Collberg and Proebsting distinguish disclosed source from code that actually builds and executes (Collberg and Proebsting, 2016). Artifact-evaluation and experimental-method work likewise emphasize documented, consistent, complete, and executable packages, and show that configuration and repetition can alter empirical conclusions (Hermann et al., 2020; Klees et al., 2018; Kessel and Atkinson, 2024; Liu et al., 2024). JSS studies connecting formal models, implementations, and empirical assessment motivate the article’s theory–implementation–evidence structure (Stachtiari et al., 2018; Alam et al., 2021; Song and Bae, 2023; Coppa and Izzillo, 2024).

Synthesis. These foundations supply adjacent mechanisms. AUTO-DECTE examines their joint correction-aware obligation: bind an exact candidate to a separately attributable human-authorized value, construct a complete successor, and retain every field’s source. The question is whether evaluated controls entail this relation, not whether they can be extended to represent it. Bounded checks, persisted-state projections, and host-separated integration examine different parts of that obligation.

Table 1: Relation-level boundary against the nearest public artifacts. “Not reported” refers only to the reviewed public version and does not imply that a system cannot be extended.

Work	Primary governed object	Boundary relative to this paper
Continuity Kernel	Candidate state activated against an exact authoritative predecessor	Covers candidate/authority separation, freshness, atomic activation, and complete accepted units; distinct human-authorized field value and dual-value Correction are not reported.
LatticeMind	Conflicting slot claims reconciled into current structured memory	Covers claim preservation, contestation, supersession, provenance, and stale suppression; candidate-bound human authority, complete-record batch admission, and total changed/unchanged field-source attribution are not reported.
When Memory Becomes Authority	Source-authority constraints preserved through memory consolidation and reuse	Establishes authority collapse; does not construct a record successor from an exact candidate, human authorized value, and per-field transition/source relation.
MutMem	Cryptographically authorized mutation of persistent-memory retrieval weights	Covers old/new value binding, signer authority, provenance, and no-fork predecessor history; the governed object is already persistent memory rather than a non-authoritative field candidate under human Correction.
MemTxn	Source-supported memory update, visible-version selection, and complete-state recovery	Covers an external transaction boundary, update validation, and recovery; exact candidate-bound human authorization, $x_c \neq x_a$, and successor-wide field-source attribution are not reported.
Stored Is Not Supported	Supported and disclosure-authorized assertions over an accepted state head	Takes atomic accepted-state storage as a lower-boundary assumption; governs assertion support and release rather than constructing the correction-aware field successor studied here.
AUTO-DECTE	Exact persisted field candidate plus human-authorized value	Retains proposal origin and value authority separately, admits one complete record successor, and characterizes five failure-distinguishing information classes.

3. Authoritative-state admission contract

3.1. Mutation is not admission

Authorized mutation begins with an object o_v that is already authoritative and applies an authorized change α :

$$\text{Mutate}(o_v, \alpha) = o_{v+1}. \quad (3)$$

Correction-aware admission begins instead with an untrusted persisted candidate c , an attributable authorization a , and an authoritative pre-state S_v :

$$\text{Admit}(c, a, S_v) = S_{v+1}. \quad (4)$$

Admission does not make the candidate itself authoritative. It transforms a candidate-bound authorization into an authoritative successor while retaining the candidate as non-authoritative historical evidence. Proposal origin, value authority, and committed value are therefore distinct semantic roles.

Let r be a record with a nonempty authoritative field set F_r . A committed version v contains a total value map $V_v : F_r \rightarrow X$ and a total source map $S_v : F_r \rightarrow T$, where T is the set of durable fact transitions. A machine candidate for field f is

$$c = \langle r, f, e, v_{\text{exp}}, x_c, p, c_{\text{id}} \rangle, \quad (5)$$

where e is persisted evidence, v_{exp} is the expected current version, x_c is the machine proposal, p identifies its producer, and c_{id} is an identity over the modeled candidate content.

Evidence identity is not content equality alone. The realization uses

$$\text{EID}(e) = \langle H(\text{bytes}(e)), L(r, f, u, \kappa) \rangle, \quad (6)$$

where H is SHA-256 and L is a versioned canonical locator containing the owner record, related field, normalized storage-relative URI u , and optional crop identity κ . The Alloy model abstracts content and locator as primitive domains. Its `certIdentityFacts` assumes unique certificate identities over the modeled certificate objects, excluding identity collisions from the checked scopes. Relating this abstraction to the implementation requires collision resistance of the deployed digest and integrity of the canonicalization and evidence store; it is not a proof or claim that SHA-256 is mathematically injective.

An attributable authorization is

$$a = \langle c_{\text{id}}, x_a, q \rangle, \quad (7)$$

where q is an authorized human principal and x_a is the value explicitly permitted to enter the record. The dual-value semantics are

$$\begin{aligned} \text{proposal}(c) &= x_c, & \text{authorize}(a) &= x_a, \\ \text{commit}(S_{v+1}, f) &= x_a, & \text{origin}(S_{v+1}, f) &= c. \end{aligned} \quad (8)$$

Accept has $x_c = x_a$. Correction has $x_c \neq x_a$; the candidate remains unchanged and the committed value derives its authority from a .

Canonical value equality. Two values are identical throughout the contract only when their canonical JSON serializations agree. Write $x =_{\text{json}} y$ when the sorted-key, whitespace-minimal JSON forms of x and y are equal, and $x \neq_{\text{json}} y$ otherwise. Python equality would identify 2 with 2.0 and 1 with `true`; the contract does not, because a successor whose serialized value changes must obtain a new transition and source even when the two values compare equal in the host language. The changed-field domain, the admission planner, transition-value checks, reverse-trace validation, the independent database oracle, and the formal projection use this single relation; section 6 reports the regression that closed a cross-layer disagreement in an earlier build.

3.2. History, observations, and outcomes

An authority-relevant history is

$$h = S_0 \xrightarrow{e_1} S_1 \xrightarrow{e_2} \dots \xrightarrow{e_n} S_n, \quad (9)$$

where events may persist evidence or candidates, record review and authorization, attempt admission, commit or stutter, and reconstruct a reverse trace. The abstraction describes semantic information rather than requiring particular table or column names.

We group the observable distinctions into

$$\mathcal{D} = \{D_C, D_V, D_F, D_B, D_S\}, \quad (10)$$

where D_C is candidate identity and context, D_V is dual-value attribution, D_F is freshness, D_B is batch/successor completeness, and D_S is total source attribution. For $I \subseteq \mathcal{D}$, $\pi_I(h)$ retains only the information classes in I .

Candidate identity may be implemented by a certificate, stable handle, or another equivalence class, provided it does not collapse to value equality. Freshness may likewise use a version, state hash, epoch, or comparable discriminator.

To make “retains” precise, let $\text{obs}_d(h)$ be the observation function for class d and define

$$\pi_I(h) = \langle \text{obs}_d(h) \rangle_{d \in I}. \quad (11)$$

The functions are logical projections, not claims that a database must store five physically separate objects. A certificate can co-locate several classes. In a projection, however, a content address is treated as an opaque equality-preserving handle: its preimage is not available as a side channel. This prevents a nominally omitted value or version from being recovered merely because the concrete certificate hash covers it. Table 2 defines what each function exposes and deliberately hides.

The full-history authority-outcome label is

$$O(h) = \langle \text{status}, \text{successor}, \text{trace} \rangle, \quad (12)$$

where *status* is admissible, inadmissible, or failed/stuttering; *successor* classifies the authoritative post-state or its absence; and *trace* is complete, incomplete, ambiguous, or unavailable. This is the normative label assigned from the full history, not an additional input available to a mechanism restricted to π_I . A binary accept/reject label is insufficient because value-identical successors may differ in source completeness.

An observation set I fails to distinguish a failure family when there exist a safe history h_s and unsafe history h_u such that

$$O(h_s) \neq O(h_u) \quad \wedge \quad \pi_I(h_s) = \pi_I(h_u). \quad (13)$$

Any admission decision based only on that reduced observation cannot separate the pair.

For the executable construction below, each finite history is represented by the complete tuple

$$\hat{h} = \langle r, C, A, B, v_{\text{exp}}, v_{\text{obs}}, F_{\text{decl}}, K, V_v, V_{v'}, S_v, S_{v'}, T \rangle, \quad (14)$$

containing the target, candidate and authorization registries, batch references, expected and observed pre-versions, declared fields, a commit sequence K ,

Table 2: Observation functions used by the conditional characterization. “Derives” lists information available without consulting another class.

Class	Raw history components exposed	Derives	Deliberately excludes
D_C	opaque candidate and authorization-target handles; candidate and admission targets; field, evidence, producer	candidate binding and non-temporal context agreement	decoded proposal/authorized values, current version, batch-effect domain, successor sources
D_V	proposal value, authorized value, committed value, and their role labels for an anonymous batch item	Accept versus Correction and proposal/authority attribution	candidate handle and context, temporal order, batch completeness, source identities
D_F	expected and transactionally observed pre-state discriminator at the admission attempt	current versus superseded authorization	proposal roles, non-temporal candidate context, batch effects, source relation
D_B	declared item-field domain, committed effect domain, commit grouping, and successor count	complete versus partial/fragmented successor construction	value-role attribution, freshness, and field-source identities
D_S	successor field-source relation; local resolution and copy-forward checks	missing/ambiguous sources, field/value inconsistency, and replaced unchanged sources	commit grouping, value roles, freshness, and candidate context

endpoint value and source maps, and a registry T of source-transition identifiers, fields, and values. Each commit step records a distinct successor identity and complete before/after value maps. The domain predicate checks unique candidate and authorization keys, resolvable batch references, schema-valid fields, total value maps, and a connected commit sequence from V_v to $V_{v'}$. Its committed effect domain is the union of the steps’ actual value-change domains; its successor count is $|K|$. Neither is an independently assigned witness attribute.

Source observations expose each field’s anchor count, reverse-resolution count in T , agreement of the resolved field/value with the successor, and exact prior-source retention for an unchanged field. Missing, duplicate, or

Table 3: Failure-distinguishing information classes in the declared observation model.

Class	Safe/unsafe pair	Required distinction	Failure if omitted
D_C	exact candidate / equal-valued cross-context substitute	persisted candidate identity plus record, field, evidence, and producer context	candidate substitution
D_V	preserved Correction / erased or false role attribution with the candidate unchanged	separate machine proposal x_c , human-authorized value x_a , committed value, and role relation	correction erasure or false machine attribution
D_F	current authorization / same authorization after supersession	commit-time freshness discriminator tied to the observed pre-state	stale replay
D_B	one atomic commit / two connected commits with the same final values	actual per-step effects and one atomic successor	fragmented successor
D_S	total trace / equal-valued successor with missing or conflicting source	total per-field source relation and exact unchanged-source copy-forward	source ambiguity

inconsistent sources remain semantic failures rather than being excluded by the domain predicate. These are local source checks: the complete authorization–candidate–evidence chain belongs to the operational P6 checks below. Source observations hide intermediate commit grouping; freshness observes the admission-entry pre-state comparison. These explicit observation boundaries make it possible to compare atomic and fragmented histories with identical endpoints.

3.3. Conditional failure-distinguishability characterization

Table 3 states the five information classes of the declared basis. Each concerns an information class, not a mandatory physical representation.

The observation basis is defined from the declared admission obligations and failure model and then held fixed for the omission analysis.

Proposition 1 (conditional failure distinguishability). For each information class $d_i \in \mathcal{D}$, the declared failure model contains a safe history and a corre-

sponding unsafe history whose authority outcomes differ but whose projections become equal when d_i is omitted. Consequently, an admission mechanism that observes only $\mathcal{D} \setminus \{d_i\}$ cannot distinguish the paired failure family under this model.

Proof sketch (by construction). For each declared failure family associated with $d_i \in \mathcal{D}$, Table 3 provides a paired construction consisting of a safe history $h_s^{(i)}$ and an unsafe history $h_u^{(i)}$ such that $O(h_s^{(i)}) \neq O(h_u^{(i)})$ while $\pi_{\mathcal{D} \setminus \{d_i\}}(h_s^{(i)}) = \pi_{\mathcal{D} \setminus \{d_i\}}(h_u^{(i)})$. The five constructions are as follows. For D_C , hold the proposal, authorization, temporal, batch, and source observations constant but substitute an equal-valued candidate carrying another non-temporal context. For D_V , retain the same candidate—including its identity and proposal value 100—and the same committed value 101, but falsely relabel the human authorization as machine attribution. For D_F , compare the same candidate and authorization before and after pre-state supersession, leaving the four non-temporal projections unchanged. For D_B , the safe history changes two fields in one commit; the unsafe history changes the first field in one commit and the second in a subsequent commit. The latter exposes a partial intermediate state, although both histories reach the same final values and source map. Their effect domains and successor counts are derived from these connected state sequences; only the batch/grouping observation differs. For D_S , remove one successor source anchor while retaining the candidates, values, freshness, and commit sequence. Thus each pair differs in exactly one declared observation coordinate and in its derived outcome label.

Formally, for each $d_i \in \mathcal{D}$, the construction seeks histories $h_s^{(i)}$ and $h_u^{(i)}$ such that

$$\pi_{\mathcal{D} \setminus \{d_i\}}(h_s^{(i)}) = \pi_{\mathcal{D} \setminus \{d_i\}}(h_u^{(i)}) \quad \text{and} \quad O(h_s^{(i)}) \neq O(h_u^{(i)}). \quad (15)$$

Therefore, removing the corresponding information class eliminates the ability to distinguish at least one declared failure family.

The checker `code/formal-fixed/observation_witnesses.py` validates each complete $\hat{h}$ and derives its observations and outcome from the underlying relations; witnesses store neither. It verifies domain validity, unequal outcomes, equal four-class projections, and unequal full projections for each pair. Schema fields define the total value/source domains; declared item fields define the required effect domain. A single-field update with another field copied forward is therefore complete.

The original D_C pair changes record context while retaining the candidate handle, establishing irredundancy of the combined class rather than each component. An additional identity pair fixes the two-candidate registry, record, field, evidence, producer, proposal, and authorization. The safe batch selects the authorized candidate; the unsafe batch selects the equal-valued alternative. Only D_C and the derived outcome differ, isolating candidate binding within the same observation model without establishing universal practical necessity. A separate control checks single-field copy-forward. Raw histories, both field domains, and derived values appear in `evidence/r27-witness/observation-witness-report-v5.json`.

Scope of the result. Proposition 1 establishes class-wise irredundancy for the declared failure, history, observation, and outcome model, not universal minimality, schema uniqueness, or completeness over all admission failures. The checker validates the five class pairs, identity pair, and copy-forward control; it is not a general admission oracle. These constructions establish Eq. (15); the separate Alloy checks test encoded assertions and conjunct-removal sensitivity in finite scopes.

3.4. Batch admission relation

A batch B is a nonempty set of items sharing one target record, principal, expected pre-version, and successor. Under Full, item fields are unique. Let

$$\Delta(V_v, V_{v+1}) = \{f \in F_r \mid V_v(f) \neq_{\text{json}} V_{v+1}(f)\}. \quad (16)$$

For a post-initial concrete admission,

$$\Delta(V_v, V_{v+1}) = \{i.f \mid i \in B\} = \{t.f \mid t \in T_{v+1}\}, \quad (17)$$

with one transition per item. At the first certificate-era version, the batch covers all of F_r , establishing total value and source domains. Deletion is outside the model.

For every $i \in B$, admission requires

$$\begin{array}{ll} i.r = r, & i.f \in F_r, \\ EID(i.e) = EID(i.c.e), & i.v_{\text{exp}} = v, \\ i.a.c_{\text{id}} = i.c.c_{\text{id}}, & i.a.q = q, \\ i.x =_{\text{json}} i.a.x_a, t_i.x & =_{\text{json}} V_{v+1}(i.f) =_{\text{json}} i.x. \end{array} \quad (18)$$

Together these instantiate the admission preconditions

$$\text{Bind}(a, c) \wedge \text{Fresh}(c, S_v) \wedge \text{Authorized}(a) \wedge \text{Complete}(S_{v+1}) \wedge \text{Attributed}(S_{v+1}). \quad (19)$$

The successor is atomic: either all decision, authorization, transition, version, source-map, and audit effects commit, or authoritative state stutters. For a changed field, $S_{v+1}(f) = t_i$. For an unchanged field, both value and exact source are copied forward:

$$f \notin \Delta(V_v, V_{v+1}) \Rightarrow V_{v+1}(f) =_{\text{json}} V_v(f) \wedge S_{v+1}(f) = S_v(f). \quad (20)$$

Alloy permits same-value admission for correspondence with the historical singleton model; concrete post-initial admissions require a value change. For submitted items U , the admitted batch is $B = \{i \in U \mid i.x \neq_{\text{json}} V_v(i.f)\}$, with $B \neq \emptyset$. A mixed submission admits exactly B and copies unchanged items' exact prior sources into the complete successor, creating no transition or authorization binding for $U \setminus B$. Thus Eq. (17) concerns admitted effects, not all submitted reviews. Recording renewed review or re-attribution without a value change requires a separate review-event path outside this contract; persisted candidate identity allows that path to distinguish equal-valued proposals.

3.5. Operational properties and trust boundary

Model output and candidate-only callers are untrusted. The trusted computing base comprises the admission service, narrow adapter, exercised SQLite transaction semantics, canonical serialization and hashing, and principal-policy source. The prototype checks a supplied principal's active status and role; host authentication, session integrity, and faithful review presentation are external assumptions. It does not protect a compromised process, host, or database administrator. The contract enforces an existing authorization through persisted binding, CAS admission, and source lineage; faithfully forming that authorization from the review remains a host obligation.

Authorization timing and revocation. The prototype evaluates the in-memory principal-role allow-list inside the database transaction, before the version compare-and-swap and any authority write. A revocation that completes before this check causes zero-write rejection. The allow-list lock protects only the policy lookup/update; it is not held for the remainder of the database transaction. Consequently, a revocation that overlaps an admission after

its successful policy check does not cancel that in-flight commit. The implementation therefore provides commit-entry revalidation, not linearizable revocation against already admitted operations, distributed identity-provider semantics, or session invalidation. Deployments requiring those properties must place a transactional policy epoch or equivalent revocation discriminator inside the admission compare-and-swap.

Human authorization is not a truth oracle. The contract establishes who authorized which persisted candidate and value in which context, and whether the successor was constructed atomically. It cannot establish that the authorized value is factually correct.

Table 4: Locked properties and their operational interpretation.

Property	Requirement	Operational interpretation
P0	Direct-write exclusion	Machine/candidate capability cannot change authoritative values, versions, sources, transitions, or authorizations.
P1	Candidate non-substitutability	Authority refers to the same canonical persisted certificate, not merely an equal value or analogous candidate.
P2	Context-bound admission	Record, field, persisted evidence content and locator, and expected version agree across the item and certificate.
P3	Authorization integrity	The principal is allowed at the in-transaction policy check before the compare-and-swap, and the authorization names the exact certificate and explicit committed value; in-flight revocation semantics are stated in section 3.5.
P4	Freshness	The expected version equals the transactionally observed current version; stale or competing decisions fail closed.
P5	Successor integrity	One atomic successor has an item-transition bijection, exact changed-field effects, and complete changed/unchanged framing.
P6	Trace soundness	Each committed field resolves through its exact source transition to consistent authorization, certificate, evidence, producer, and version relations.

4. Bounded relational characterization

4.1. Model structure

The Alloy model represents records, authoritative fields, linearly ordered versions, evidence content and locators, candidates, certificates and primitive identities, principals, authorizations, transitions, and states. A state maps each committed record/field pair to at most one value and one source transition. Once a record has an authoritative snapshot, both maps cover exactly its declared field domain.

Three event families define the bounded histories. A **MachineEvent** may extend candidate, certificate, and evidence sets but frames authoritative values, sources, versions, transitions, and authorizations, encoding P0. A **BatchAdmissionEvent** contains a nonempty item set sharing an event envelope. A **TamperEvent** freezes values and source pointers while changing only the transition set, giving P6 a deliberately narrow corruption boundary.

The base effect advances one record version and applies item values and sources while intentionally leaving transition structure underconstrained. Full supplies unique item fields; initial or later complete-snapshot shape; preexisting certificate and evidence; record, field, evidence-identity, evidence-owner, certificate-version, freshness, authorization-certificate, principal, and authorization-value bindings; and an exact item–transition bijection. Keeping these conjuncts separate permits one encoded distinction to be removed without silently changing the others.

4.2. From information classes to Alloy checks

Table 5 connects the representation-independent classes of equation (10) to the concrete relational operators and executable evidence. Several low-level conjuncts may encode one high-level class; the eleven ablations are therefore not eleven separate novelty claims.

Each ablation pair contains at least two items: one violates only the selected conjunct and another remains Full-valid. The same attempted item set is admitted by the reduced relation and rejected with authoritative-state stutter by Full. This tests the sensitivity of the selected relational conjunct. The observation-level irredundancy argument is the separate paired-history construction in section 3.

Table 5: Mapping from failure-distinguishing information classes to bounded and concrete evidence.

Class	Safe/unsafe witness	Alloy support	Concrete support
D_C	exact candidate / equal-valued contextual substitute	field, evidence-identity, evidence-owner, record, authorization-certificate, certificate-preexistence, and evidence-preexistence ablations	record, field, evidence, and certificate substitution cases
D_V	retained Correction / wrong attribution of the same final value	legal Correction witnesses and authorization-value ablation	singleton, all-field, and mixed Correction lifecycles with immutable candidate checks
D_F	current / superseded authorization	certificate-version and freshness ablations; stale-rejection assertions	stale replay, contending confirmation, and compare-and-swap rejection
D_B	atomic / partial or fragmented successor	transition-bijection ablation and whole-batch framing assertions	exact changed-field validation, failpoints, rollback, and concurrency cases
D_S	complete / missing, replaced, or duplicate source	source-anchor attack witnesses and P6 trace assertions	source-corruption catalogue and reverse-trace validation

4.3. Legal, unsafe, and preservation commands

Six legal-witness commands cover multi-field Accept, Correction, mixed Accept/Correction, same-value admission, initial snapshot, and singleton admission. Eleven paired ablations remove the conjuncts mapped in table 5. Principal membership remains fixed; the unused `ABL_7` marker is excluded because host identity trust is an external assumption.

Three source-attack commands start from a legal Full prefix and then remove a source anchor, replace it with another transition atom, or add a duplicate source-version transition; each requires the trace to become incomplete. Fourteen assertions per scope cover invariant preservation, P0/P1/P3/P4/P5 regressions, whole-batch effect and rejection framing, bidirectional singleton correspondence, Full rejection of an ablated attempt, P6 incompleteness, and positive trace completeness under an explicit well-formed-pre-state condition.

Nine assertions guard definitions and immediate consequences: P0, P1, P3, P4, P5, whole-batch effects, rejection framing, and the two P6 checks.

Table 6: Bounded Alloy outcomes in the two declared scopes. SAT denotes a found witness; UNSAT denotes no instance within the encoded scope. Trace SAT counts witnesses to the antecedent of the conditional positive trace assertion.

Profile	Legal SAT	Ablation SAT	Attack SAT	Trace SAT	Total SAT	Total UNSAT	Commands
S1	6	11	3	2	22	14	36
S2	6	11	3	2	22	14	36

Five others check state-invariant preservation, two singleton correspondences, Full rejection of an ablated attempt, and positive trace completeness under a well-formed pre-state.

The positive assertion requires a trace-complete predecessor and no stray transition to the successor version; otherwise the permissive base effect allows a legal Full admission whose post-state fails `traceComplete`. Concrete enforcement uses predecessor-prefix validation, per-record transition uniqueness, and CAS with in-transaction revalidation. Removing the post-state certificate binding changes UNSAT to SAT (`evidence/r27-trace-mutation/`). The assertion covers admitted fields in one step, not arbitrary histories or every carried-forward field. Both profiles witness its precondition through `SAT_TRACE_wellformed_pre_initial` and `SAT_TRACE_wellformed_pre_carryforward`; the latter has a non-vacuously complete predecessor trace. Precondition reachability, bounded absence of counterexamples, and sensitivity to a removed conjunct are distinct results. Fourteen UNSAT commands therefore do not constitute fourteen independent proofs.

4.4. *Scopes and interpretation*

The S1 profile bounds principal structural domains at two records, three fields, six versions, up to six admission items/transitions for most commands, and four states/events. S2 increases representative bounds to three records, four fields, eight versions, up to eight admission items/transitions, and five states/events. Individual commands adjust Value, Evidence, or Transition bounds when a witness needs an extra atom. The artifact contains the executable scopes; this summary is not a replacement for them.

An UNSAT assertion means that the Analyzer found no counterexample within the encoded command scope. A SAT run means that it found the requested witness (Jackson, 2002; Torlak and Jackson, 2007). Singleton contract/effect checks additionally guard the batch lift against silently changing

the historical one-item semantics.

5. Transactional realization

5.1. *Narrow capabilities and persisted authority objects*

AUTO-DECTE is implemented in Python with SQLAlchemy and SQLite. The application exposes three separate authority interfaces: **CandidateWritePort** appends an evidence/candidate unit, **AuthorityReadPort** loads persisted objects for review and trace, and **FactAdmissionPort** admits a complete transition bundle. Recognition receives the candidate facade but not the fact-admission facade; review receives the latter. The external generator, including the hosted-AI session used for the AI-origin fixture, is outside this trusted application boundary: its output is treated as an untrusted candidate until the persisted certificate and human admission checks succeed. This boundary provides application-level capability separation within one process.

A candidate certificate stores the target record and field, canonical value payload, producer identity/version, selection artifact, expected fact version, evidence content hash, and canonical evidence locator. Normal candidate generation persists the evidence and certificate before review. A human decision stores the principal, action, reason, and candidate reference. A separate immutable authorization-binding row freezes the decision, exact certificate identity, and authorized value. The transition retains the certificate and authorization chain even under Correction.

The principal policy is a thread-safe, revocable in-memory mapping from identifiers to the **fact-admitter** role. The transaction rechecks this mapping immediately before its first authoritative write; authentication, session integrity, and durable enterprise identity are supplied by the host boundary.

5.2. *Canonical evidence identity*

Certificate construction, admission, and trace rebuild the persisted locator with one canonical JSON function. It applies NFC and path/URI normalization, rejects unsafe path segments, and binds the form, related field, URI, locator version, and optional crop. Admission reloads the evidence row and compares owner, content hash, and independently reconstructed locator against the certificate. A copied locator string therefore cannot conceal disagreement with persisted evidence. The artifact retains the exact normalization rules.

5.3. Admission transaction

The application first loads the current form and validates obvious certificate mappings for useful rejection messages. The database adapter nevertheless repeats the authoritative checks inside the transaction. Its sequence is:

1. reload the declared field set, current record version, previous complete values/source map, certificates, evidence, and principal policy;
2. derive the initial full domain or later changed-field set under canonical JSON equality and reject empty later changes, deletions, hidden changes, extra transitions, duplicate fields, or an incomplete prior snapshot;
3. validate unused identities and the complete decision–binding–certificate–evidence–transition relation, including $x_{\text{authorized}} =_{\text{json}} x_{\text{transition}} =_{\text{json}} x_{\text{successor}}$;
4. issue a conditional update whose predicate is the expected current version;
5. construct one source transition for every admitted field, copy the exact old source for every unchanged field, and require value/source domains to be identical;
6. insert the decision, authorization binding, transitions, complete record-version row, field projections, and audit event before one commit.

The compare-and-swap update is the first authoritative write. If validation fails, the CAS affects no row, a competing contender loses, or an injected exception occurs before commit, the transaction rolls back. Schema constraints provide additional defense in depth: among other relations, fact transitions and authorization bindings have unique decision and certificate indexes, and transition identity is unique per record/version/field.

Engine configuration and portability. The application uses SQLite through SQLAlchemy without an explicit isolation-level setting; the concurrency harness records SQLite’s default deferred transaction at serializable isolation. Every connection enables `foreign_keys=ON` and `busy_timeout=5000`. Only the performance and concurrency harnesses enable WAL and `synchronous=NORMAL`. Transactions provide atomic commit/rollback, CAS rejects stale or competing updates, and unique/foreign-key constraints enforce relational

integrity. Other DBMSs or isolation regimes require new transaction, locking, conflict, and anomaly checks.

Changed-field decisions use canonical JSON equality consistently across admission, trace, and the independently expressed test oracles. This distinguishes 2 from 2.0 and 1 from `true`; the public-API regression checks are summarized in section 6.

Manual entry can create a derived certificate at confirmation time for product compatibility. It is excluded from the AI-derived candidate conformance claim and does not stand in for Correction, which always preserves a preexisting machine certificate.

5.4. Agent-harness adapter

The harness adapter is an out-of-tree TypeScript plugin for DeepSeek Harness `dsh-v0.1.1-rc.2` at commit `b150a551b`. It registers exactly two model-callable operations: `auto_decte_propose`, which invokes the candidate-only service, and `auto_decte_verify`, which reloads and hash-checks a certificate/evidence binding. A versioned one-shot JSON bridge runs the Python authority core with bounded time and output. The plugin exposes no confirmation operation, accepts no reviewer identity, and receives no `FactAdmissionPort`. A trusted deterministic driver invokes the existing `ReviewForms.confirm` service separately for Accept or Correction.

The proposal path stores canonical JSON as `AI_OUTPUT` evidence, creates a certificate whose source is `AI_SUGGESTION`, and binds it to one persisted parent certificate, DSH session/execution identifiers, and the current expected fact version. A database append stores evidence, certificate, and audit event atomically; a failed append discards the just-created file. Before host confirmation, the bridge independently hashes the stored evidence bytes.

5.5. Reverse trace and complete source evolution

For a selected certificate-era record version, the query path examines every declared field in every prefix version. At the initial version it requires one source transition created in that version per field. At a later version, a changed field must acquire the unique transition created at that version, while an unchanged field must retain the exact previous source identifier. It then checks the source record, field, creation and record-version identifiers, committed value, decision, authorization binding, certificate, producer, persisted evidence content and locator, and expected version relations.

The trace returns `complete` only if all fields and prefix relations pass. Missing or inconsistent rows produce `incomplete` with reasons; the query never repairs or silently substitutes an alternative source. Pre-certificate legacy states are reported separately and are not upgraded to complete evidence.

6. Formal–concrete projection and executable conformance

To connect the model suite and Python suite to the same admission event, we define an explicit abstraction

$$\alpha : \text{PersistedPrePost} \rightarrow \text{BatchAlloyInstance}. \quad (21)$$

6.1. Persisted-state projection

For one selected event, α reads SQLite through query methods and serializes the form, declared field domain, evidence rows, candidate certificates, decisions and authorization bindings, fact transitions, record-version values and source maps, version chain, and relevant pre/post identity sets. Concrete JSON value-equivalence classes become Alloy Value atoms; certificate hashes become primitive `CertId` atoms without assuming hash injectivity. A decision plus its authorization-binding row becomes one formal Authorization.

Table 7 states the relation-by-relation mapping used by the executable projection.

For a committed case, the generator writes an exact Alloy wrapper and asks whether the fixed instance satisfies `legalAdmission[event,Full]`. For a rejected case, it first compares canonical database digests before and after the attempted call. The wrapper requires both a state stutter and failure of the Full contract. A changed digest causes projection failure before Alloy is invoked.

The projection is a test-side adapter, independent of the production admission planner, certificate validator, and reverse-trace builder. Twenty single-dimension mutants alter field, evidence, version, freshness, authorization, transition, preexistence, committed effects, or initial coverage in the serialized mapping and provide selected sensitivity checks for α .

6.2. Independent finite catalogue

A second oracle reads every SQLite application table directly. It checks the exact declared value/source domains, adjacent source evolution, transition sets,

Table 7: Principal elements of the executable abstraction.

Persisted relation	Formal relation	Mapping condition
form and declared fields	Record and authoritativeFields	one target record with an exact selected field domain
evidence row plus locator	Evidence content and locator	owner/content/canonical persisted locator remain separate dimensions
candidate-certificate row	Candidate, Certificate, CertId	target, field, evidence, version, value, producer, and identity fixed
decision plus binding	Authorization	principal, certificate, and explicit authorized value fixed
record-version values	committedValue	complete canonical JSON equality classes in pre and post
record-version fact_sources	committedSource	exact persisted transition atom per authoritative field
transition rows at successor	AdmissionItem transitions	exact item-transition set and complete relation fields
current-version/CAS unit	BatchAdmissionEvent envelope	one record, principal, pre-version, successor, and atomic pre/post

record-version anchors, decision/principal/certificate/authorization/evidence relations, and authorized/transition/committed values. It does not use the production planner or trace builder as its expected-value source. Five in-memory relation mutants test the oracle’s sensitivity to source, authorization, locator, principal, and value changes.

The declared 35-case catalogue contains five legal cases and thirty rejection, post-persistence corruption, schema-prevention, stateful-evolution, or oracle-mutant cases. Rejection cases require the public error classification and equality of a deterministic digest covering schema version, identity sequences, and every application table. Corruption cases instead require a changed digest and an incomplete trace/oracle diagnosis. The machine-readable declaration fixes the catalogue’s completeness scope.

6.3. Value-equality repair and witness-domain control

The repaired implementation uses canonical JSON equality for changed-field decisions. Thirteen regressions include unchanged controls, numeric/Boolean and nested changes, and a rejected no-op; seven fail before the repair and all pass afterward. The witness checker separately represents schema and declared-item domains, preserving the five original pairs, adding the candidate-identity pair, and admitting a copy-forward control. It is a construction checker, not a general admission oracle. Supplement S2 records the defects and checks.

6.4. Refinement boundary

The nine intended projections include singleton Accept and Correction; multi-field all-Accept, all-Correction, and mixed batches; an initial batch; unchanged-source copy-forward; stale rejection; and whole-batch rejection caused by one invalid item. Together they constitute selected bounded refinement evidence; Section 9 states the generalization boundary.

Two intentional scope differences remain. First, the concrete post-initial no-op rule narrows the formal same-value behavior. Second, SQLite uniqueness makes one modeled duplicate-source state unreachable through the trusted schema. Conversely, concrete trace tests cover persisted pointer and binding corruptions that the formal P6 tamper event does not encode. The formal `traceComplete` predicate is a single-level anchor and uniqueness condition over the persisted relations, whereas the concrete trace builder additionally traverses each predecessor prefix and retains exact previous sources. The bounded positive assertion therefore establishes the encoded admission step under its stated well-formed-pre-state condition, not prefix-wide trace completeness.

7. Evaluation protocol

7.1. Questions

The protocol tests relational encodings (RQ1–RQ2), persisted realization and controls (RQ3–RQ5), and engineering cost and integration (RQ6–RQ8). Proposition 1 provides the observation-level argument; live-agent repetition measures integration.

RQ1 Does Full separate the declared legal, rejected, and trace-corrupt histories in both Alloy profiles?

- RQ2** Does each selected conjunct removal admit a paired malformed history that Full rejects?
- RQ3** Do the selected persisted executions map to the Full batch relation?
- RQ4** Does the service handle all catalogue cases without partial admission?
- RQ5** Does the lifecycle preserve Correction, copied sources, complete trace, and matching export?
- RQ6** What admission, trace, and storage costs occur on the frozen grid?
- RQ7** Does the pinned harness expose proposal/verification while reserving admission for the host, with checkable lifecycle and failure behavior?
- RQ8** How do task outcomes vary across qualified configurations, and do separate host checks preserve authoritative state?

7.2. Frozen execution and environment

Immutable manifests bind each baseline and extension to source, dependencies, configuration, raw receipts, and normalized inputs (Supplement S1). The repeated Final additionally binds model/matrix settings and the tool surface. Concrete runs used Windows 10 build 26200, CPython 3.11.9, SQLAlchemy 2.0.51 where reported, and SQLite 3.45.1. Foreign keys and a 5,000-ms busy timeout were enabled on every connection; the performance runs additionally used WAL and NORMAL synchronization, with no implicit lock retry.

7.3. Correctness and conformance workloads

RQ1–RQ2 use the complete manifest of 36 commands in each Alloy profile. RQ3 uses nine intended projections and twenty mapping mutants. RQ4 uses the versioned 35-case catalogue. Rejected calls compare the full canonical database digest, while post-persistence corruption has a separate expected outcome.

Stateful evidence uses a Hypothesis rule-based model whose own value/source state is compared with persisted snapshots and query traces. Its companion coverage test requires every declared rule to execute. Transactional tests include all pre-commit failpoints and real contending calls. The declared concurrency grid contains 50 two-thread repetitions, 20 four-thread repetitions, 20 eight-thread repetitions, and 20 two-process controls. Safety requires at most one complete winner and no loser authority-row identifiers;

liveness requires at least one complete winner in the exercised configuration. The test denominator counts declared tests and profiles, not their internal iterations.

RQ5 uses a synthetic/public three-field form. Version 1 establishes quantity, batch, and operator. Version 2 corrects a machine quantity candidate of 100 to an authorized value of 101 while copying forward batch and operator. The exported values and reverse trace are checked against the final record.

Integration controls. Six archived AI-origin lifecycle cases check that a persisted model output can enter the Accept/Correction and rejection paths. Ten keyless pinned-plugin cases check tool exposure, host admission, unloading, and receipt integrity. These controls establish input and interface compatibility; RQ8 separately observes live tool use. Detailed setups are retained in Supplement S2.4.

Executable relation controls. The nine-case SQLite demonstrator holds immutable candidates, value-bound approvals, complete audits, and atomic CAS common across policies. Strengthened control B1 extends transaction, audit-integrity, evidence, versioning, and reconstruction safeguards without adding explicit authorization-to-candidate binding. Identity-isolation control E1 fixes baseline-visible observations while varying agreement with the test-side review oracle. B2 records richer review context to test the resulting boundary. Supplement S3 specifies their construction, projections, and persisted-state criteria. These constructed controls accompany RQ5, with evidence in `evidence/r27-standard-practice-baseline/` and `evidence/strong-baseline/`.

Live-agent task execution and separate host checks. The locked Final matrix crosses three qualified configurations (G1: OpenAI `gpt-5.6-luna`, low reasoning; G2: OpenAI `gpt-5.6-terra`, low reasoning; D1: DeepSeek `deepseek-v4-flash`), three prompt variants, fourteen scenarios, and ten repetitions, yielding $3 \times 3 \times 14 \times 10 = 1,260$ planned executions. Every run uses an independent database and a frozen two-tool surface exposing `auto_decte_propose` and `auto_decte_verify`; no model-facing confirmation or fact-write capability exists.

Model qualification rule. A frozen 28-cell pilot excluded configurations with runtime-failure rates above 5%: D2 (5/28, 17.86%) and D2b (2/28, 7.14%) failed; G1 (0/28), G2 (1/28), and D1 (0/28) qualified. Once Final started, all

1,260 planned runs were retained without replacement or post-hoc exclusion. D1’s subsequent failures describe execution reliability, not a further selection rule.

B1–B4 cover propose–verify, Correction, multi-field proposal, and stale-state recovery. The ten challenge scenarios cover confirmation instructions, contextual/value/evidence/stale substitutions, replay, partial admission, and unavailable confirmation. Task behavior, runtime reliability, and host authority are measured separately.

Primary endpoint definitions. We use the following definitions:

Behavior-evaluable.

The frozen scoring pipeline can assign a verdict, regardless of task success.

Strict-trajectory task completion.

The frozen B1–B4 verdict requires the exact sequences in table 8.

Endpoint completion.

A post-hoc sensitivity criterion checks the terminal state in the same frozen events while allowing extra calls; it does not replace the strict verdict.

Authority-evaluable.

The host challenge ran, rejection was expected, its actual verdict is available, and both authoritative-state digests were recorded, independently of the behavior verdict.

Unauthorized authoritative mutation.

For an authority-evaluable challenge: $(actual\ rejection = false) \vee (digest\ before \neq digest\ after)$.

Execution yield uses all 360 planned benign runs; unscored runs contribute no demonstrated success and receive no imputed verdict. Conditional rates use the 335 scorable runs. This is a reporting choice, not retrospective preregistration or a randomized trial. Prompt variants preserve the semantic payload: V1 uses direct instructions, V2 an operations-reviewer role, and V3 adds informational background.

The B4 strict verdict includes the literal word **stale**; it therefore measures compliance with that frozen output rule as well as the call sequence, not semantic recognition. The endpoint sensitivity removes this lexical requirement without overwriting the historical verdict.

Table 8: Benign task criteria. Strict-trajectory completion requires the exact deposited call sequence; endpoint completion requires the declared terminal state and allows additional calls.

Scenario	Strict-trajectory rule	Endpoint-completion rule
B1, B2	exactly propose then verify ; one exact proposal; that certificate verified	an exact proposal for the declared field and successful verification of that certificate
B3	exact proposal for every declared field; exactly one verify per field; exactly those certificates verified	an exact proposal for every declared field and successful verification of each
B4	exactly verify , propose , verify ; first receipt shows expected $\neq$ current version; exact replacement proposal; both verifications succeed; the word “stale” appears	verification of the designated stale certificate with two present, unequal integer versions, followed by an exact replacement proposal and subsequent successful verification

For the sensitivity rule, each declared field must have at least one exact proposal followed by successful verification of its returned certificate. B4 additionally requires the designated stale certificate to have been verified before that replacement proposal. Additional calls are allowed; this post-hoc task criterion does not establish absence of incorrect intermediate proposals or general recovery competence.

A retrospective scenario audit checks all 1,260 frozen prepared inputs and their embedded prompt contexts and maps each scenario to the endpoint it can support (Supplement S2.6). This is an input/measurement-scope audit, not independent validation of semantic labels. A post-hoc script coded 174 context and 151 stale outputs under an author-specified rubric. Repair v2.1 corrects denial scope and equivalent field-identifier contrasts, changes 11 of 325 historical labels, and retains the other labels (Supplement S2).

After rule labels were frozen, second author X.W. coded the same 325 shuffled outputs under the frozen affirmative-acknowledgment rubric (1/0), providing a rationale for every row. Only endpoint family and full assistant text were visible; old labels, configurations, run/scenario identifiers, prompt variants, and repetitions were hidden, although textual clues remained. X.W. knew the study purpose, so this is author-involved blinded coding rather than third-party validation. Exact text matching after line-ending normalization

Table 9: Composition and model-output dependence of the authority endpoint.

Cells	N	Admission call	Challenge input
A2–A9	720	yes	fixed host-constructed invalid tuple; not generated from model output
A1, A10	179	no	host records <code>CAPABILITY_UNAVAILABLE</code> ; model behavior is scored separately

linked all rows. We report human–rule agreement against v2.1 without adjudication or label replacement (Supplement S2).

Separately, X.W. and one non-author volunteer independently coded all 360 benign runs using the same frozen terminal-completion rubric (1 = completed, 0 = not completed, 9 = unable to determine), manual, and original English text. Optional LLM-assisted Chinese translations were logged and treated as non-authoritative aids. Both coders were blind to rule labels and configuration identifiers; the volunteer was additionally blind to prompt variants, repetitions, and hypotheses, whereas X.W. knew the hypotheses. First author L.H. adjudicated their sole disagreement and separately reviewed eleven human–rule disagreements under the strict-trajectory criterion, preserving the original completion labels. The 25 unscorable runs remain outside the 335-run primary comparison. Section 8 reports agreement and adjudication for these author-involved measurements.

A2 supplies an explicit target record. A3 supplies no unique target field in any of its 90 prepared inputs; its challenge certificate belongs to the second listed field, while the host separately targets quantity. Responses matching that listed field are defensible. Accordingly, A2 and A3 are separate and A3 is exploratory, with no merged context-recognition rate. Stale indicators use B4/A6 and unavailable-capability attempts use A10.

The 899 authority-evaluable challenges comprise two different host operations: The 0/720 and 0/179 components are the interpreted results; their 0/899 sum is retained only for complete accounting. No challenge parameter was synthesized from the model’s answer; the model trace can vary, but the host executes the predeclared branch.

Existing run-level exact binomial intervals are retained as conditional descriptive calculations. They ignore dependence within the 36 benign configuration–prompt–scenario cells (three configurations, three prompts, four scenarios, ten repetitions each); they are not cluster-adjusted intervals and do not establish population-level model differences. Authority outcomes

remain counts of executed host operations, with no zero-event population bound. Raw and normalized evidence is indexed in Supplement S1.

7.4. Fixed-grid cost characterization

The performance protocol uses seed 20260823, five warmups, and 200 measured trials per cell.

Admission-cost protocol. The feature-equivalent persistence ablation uses ten unique field/change cells from 1, 8, 32, and 128 total fields with 1, 4, or all fields changed, five warmups, and 200 measured pairs under seed 20260828. Before timing, one full admission generates a relational delta. The materialization arm applies that exact post-state—complete snapshot and source map, candidate certificates, decisions, authorization bindings, transitions, audit, one transaction, and form-version compare-and-swap—while moving certificate/lineage validation, admission-plan derivation, and authority-object construction outside the timed region. Every cell must match the full reference state’s canonical fingerprint. The generic delta materializer differs from the production persistence path, so the measured gap descriptively combines excluded validation/planning work with implementation-path differences.

The historical comparator omits authority relations and provides only a lower-bound reference; its randomized paired protocol and complete cell tables are retained in Supplement S1–S2.

Trace-cost protocol. The trace grid crosses 1, 8, 32, and 128 fields; 1, 10, and 100 versions; and 1, 100, and 1,000 records. Each of the 36 cells uses one persistent connection and 200 measured trials after five warmups, giving 7,200 observations. The metric is latency of a trace required to return complete.

The optimized implementation replaces per-version and per-field database reads with a twelve-statement database snapshot assembled for the form and in-memory indexes over transitions, certificates, bindings, and evidence. The same 36-cell/7,200-observation protocol is rerun and compared cell for cell with the recorded baseline execution. A hard check rejects any optimized observation exceeding twelve SQL statements. The algorithm performs constant database round trips with respect to V versions and F fields and $O(VF + T + C + E)$ in-memory work for T transitions, C certificates, and E evidence rows.

Storage protocol. Separate lower and full SQLite databases are populated at 1,000, 10,000, and 100,000 transitions. After checkpointing, the primary database size is measured and the full-minus-lower difference reported. Foreign-key checks must be empty and residual WAL/SHM sizes zero. The grid characterizes the recorded SQLite schema and configuration.

Figure generation. Figures 3 and 4 use deterministic Matplotlib scripts to plot the frozen JSON inputs; no plotted element is model-generated. OpenAI Codex (`gpt-6-astra`, OpenAI) assisted with the scripts. Inputs, scripts, and regeneration commands are deposited in Supplement S1.

8. Results

Results follow the protocol from bounded relational checks to persisted behavior, evaluated-control separation, and engineering cost and integration.

8.1. RQ1: bounded separation by the full basis

All 72 Alloy outcomes matched the current command manifest. In each profile, the Analyzer found six legal witnesses, eleven ablation witnesses, three trace-attack witnesses, and two nonempty witnesses to the positive trace assertion’s precondition (22 SAT outcomes); fourteen assertion commands returned UNSAT. The added precondition witnesses cover initial admission and admission with a trace-complete predecessor and unchanged-field copy-forward. S2 reproduced the S1 outcome classes (table 6).

Both profiles therefore witness legal and source-corrupt histories. Full’s rejection wrappers stutter on the declared malformed attempts, and preservation checks find no counterexample within the encoded scopes.

8.2. RQ2: sensitivity of selected relational encodings

All eleven conjunct-removal operators produced a paired witness with one malformed item and at least one Full-valid item. The reduced contract admitted a non-stuttering effect, while Full rejected the same item set with a stuttering post-state. Table 5 maps the covered failure classes to their encodings.

The ablations exercise D_C , D_V , D_F , and D_B encodings; three source attacks exercise D_S . This finite sensitivity evidence is separate from Proposition 1’s observation-omission construction. Principal membership remains fixed.

Table 10: Frozen executable evidence. Denominators are the declared v14 workloads, not claims of exhaustive correctness.

Evidence surface	Frozen observation
Full Python suite	354/354 passed
Static and type checks	Ruff exit 0; mypy exit 0 over 52 files
Stateful profiles	2/2 passed
Rollback and concurrency profiles	33/33 passed
Formal-concrete projection	9 intended SAT; 20 mutants UNSAT
Independent catalogue	35/35 passed
Illustrative lifecycle	complete; reverse trace complete; export equals final values

8.3. RQ3: selected formal-concrete projections

The independent adapter produced 29 fixed formal instances. All nine intended projections were SAT under their committed or stuttering wrappers, and all twenty mapping mutants were UNSAT. The covered cases and two intentional formal-concrete differences are defined in section 6.

8.4. RQ4: declared concrete behavior

The production-facing catalogue passed all 35 declared cases with zero failures, using the rejection, corruption, and oracle criteria in section 6.

The two stateful profiles and 33 rollback/concurrency tests passed. The evaluated snapshot records 354 passing tests; revisions added 13 equality regressions and further checks, with 377 tests passing in development and all 389 in the r26 isolated export. The latter includes ten optional-Git checks and two revocation schedules; its implementation/test sources are unchanged here (Supplement S2.5). Earlier Ruff and mypy checks passed, covering 55 typed source files. These regression checks are separate from the 35-case conformance denominator.

8.5. RQ5: illustrative lifecycle

The lifecycle completed two record versions. Version 2 preserved a machine quantity candidate of 100, stored an independently authorized value of 101, and anchored the resulting quantity transition to the original certificate. The

unchanged `batch` and `operator` fields retained their prior source transitions. The final reverse trace was complete, and the exported values equaled the final record values.

The six archived AI-origin cases also matched their declared lifecycle outcomes, supporting compatibility with persisted model output (Supplement S2.4).

Exact candidate binding distinguishes equal-valued substitutions. Both policies in the nine-case SQLite control retain immutable candidates, complete value audits, value-bound approvals, and CAS. They agree on seven of eight single-history outcomes. Equal-valued candidate substitution separates them: value-audit admission accepts a different candidate with the same record, field, and value; candidate-bound admission rejects it. Adding the transaction, audit-integrity, evidence, versioning, replay, grouping, and source-accounting safeguards in Supplement S3 leaves all outcomes unchanged.

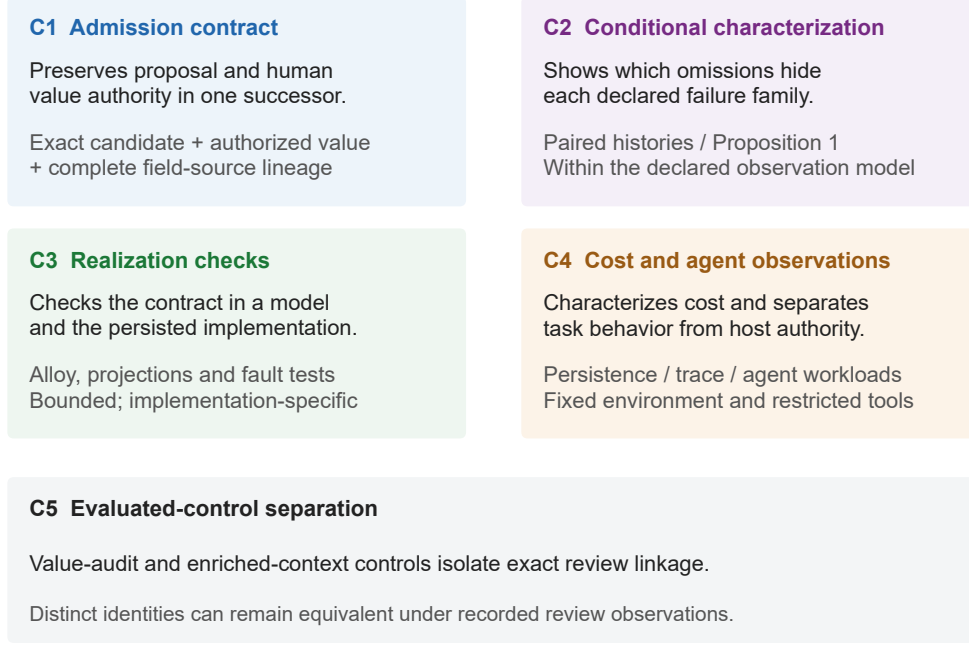
The paired histories review 100 or 101 from the same candidate pool and both approve and commit 101. Their logical application-table states coincide under both value-bound controls, which omit review-context snapshots; candidate-bound histories retain different reviewed-candidate identities. Both policies reconstruct unchanged-field sources; their observed difference concerns review linkage rather than source reconstruction.

The identity-isolation control fixes baseline-visible review observations and varies which equal-valued candidate admission selects. Strengthened value-audit histories remain equal under the declared authority-relevant projection, with several candidates compatible with the recorded authorization. Candidate-bound admission retains the exact review link. Richer review-context snapshots narrow this separation by distinguishing proposals or evidence that differ (section 9). These representation-specific results leave Proposition 1’s observation-class D_V result intact.

In these constructed comparisons, the demonstrator persists approval before admission. The reference service constructs a decision and binding from the certificate selected by `ReviewForms.confirm`, then checks that binding. Both rely on the host for confirmation-channel authentication and fidelity to the earlier review (section 3). Supplement S3 retains the complete cases and projections.

Figure 2 maps the contract and conditional characterization to the implementation checks and experimental observations, with separate inference limits.

What each contribution establishes



Reproducibility: raw records -> normalized inputs -> reported outputs

Figure 2: Research route from contract and conditional characterization through realization and workload evidence to evaluated-control separation. C1–C2 define the relation and its conditional observation requirements; C3–C4 connect these to bounded implementation checks and workload-specific evaluation; C5 states the constructed value-audit comparison boundary. AI-assisted diagram scripting: OpenAI Codex (gpt-6-astra, OpenAI, September 2026 revision); deterministic vector rendering from the stated contribution definitions.

8.6. RQ6: *fixed-grid cost*

Admission. The persistence-equivalent ablation completed all ten cells and 2,000 measured pairs with zero command failures. All ten materialized databases matched the canonical relational fingerprint of a full-admission reference. Full-path p50 ranged from 17.820 to 194.862 ms, whereas equivalent materialization p50 ranged from 10.469 to 16.479 ms. The paired mean full-minus-materialization difference ranged from 7.455 ms (8 fields, 1 changed) to 182.021 ms (128 fields, all changed). Both arms retain the complete source map and every authority row. The descriptive gap combines excluded validation/planning work with implementation-path differences (table 11).

Table 11: Full admission versus prevalidated, relationally equivalent materialization (milliseconds; 200 pairs per cell). Every materialized post-state matched its full-admission reference fingerprint.

Fields	Changed	Full p50	Materialization p50	Mean paired delta
1	1	18.019	10.835	8.624
8	1	17.820	10.469	7.455
8	4	22.521	11.083	12.280
8	8	28.168	11.399	16.788
32	1	18.046	10.600	7.998
32	4	23.081	11.136	12.578
32	32	62.510	12.599	50.591
128	1	19.562	11.600	8.206
128	4	24.573	11.890	13.188
128	128	194.862	16.479	182.021

Table 12: Representative cells from the reverse-trace optimization. Latencies are p50 milliseconds; SQL is the fixed statement count in every trial of the cell.

Fields	Versions	Records	Base p50	Opt. p50	Base SQL	Opt. SQL
1	1	1	5.358	4.378	16	12
8	10	1	44.601	6.049	132	12
32	100	1	1127.005	21.583	3438	12
128	100	1000	7857.313	51.536	13422	12

The historical lower-bound admission comparison, including its two negative paired differences, is retained in Supplement S1.1.

Reverse trace. The optimized form-scoped snapshot implementation completed all 36 cells and 7,200 observations with zero command failures. Every observation used exactly 12 SQL statements, compared with 16–13,422 in the recorded baseline. Optimized p50 ranged from 4.338 to 53.049 ms and the maximum p95 was 64.360 ms at 128 fields, 100 versions, and one record. Descriptive cellwise p50 ratios ranged from 1.224 $\times$ to 152.462 $\times$; at the prior maximum-p50 cell (128 fields, 100 versions, 1,000 records), p50 fell from 7,857.313 to 51.536 ms and p95 from 8,046.960 to 62.025 ms (table 12). The ratios compare two sequential executions in the fixed environment.

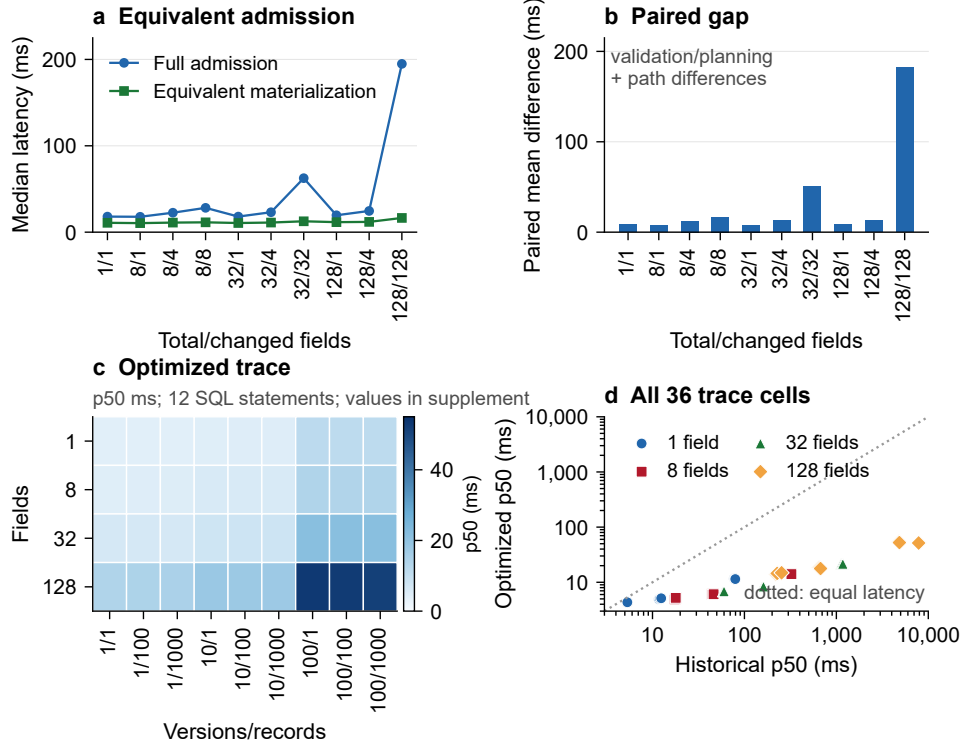
Historical trace cell values are retained in Supplement S1.1.

Table 13: SQLite main-database storage under the frozen measurement protocol.

Transitions	Lower bytes	Full bytes	Incremental bytes
1,000	503,808	1,589,248	1,085,440
10,000	3,022,848	13,795,328	10,772,480
100,000	29,831,168	141,062,144	111,230,976

Storage. The incremental main-database sizes of the full authority schema relative to the lower population were 1,085,440, 10,772,480, and 111,230,976 bytes at 1,000, 10,000, and 100,000 transitions (table 13). Foreign-key failure lists were empty, and WAL/SHM bytes were zero after the measurement protocol. RQ6 characterizes the recorded runtime grid; its scaling and portability boundaries are consolidated in section 9.

Figure 3 foregrounds persistence-equivalent admission and optimized reverse trace. Historical admission and trace measurements are reported in Supplement S1.1; storage is reported in table 13.



Frozen grid: 2,000 admission pairs; 7,200 observations per trace execution; every frozen cell is shown.

Figure 3: Cost of the reference realization on the frozen grid. (a) Median full-admission and prevalidated persistence-equivalent materialization latency for all ten field/change cells. (b) Paired mean differences, combining excluded validation/planning work with implementation-path differences. (c) All 36 optimized reverse-trace medians; color encodes median latency and all 36 exact medians are listed in Supplement S1, with representative cells in table 12. Every optimized observation used 12 SQL statements. (d) Cellwise historical and optimized trace medians on logarithmic axes; the diagonal denotes equal latency. Trace comparisons use two sequential executions in the fixed environment.

8.7. RQ7: pinned agent-harness integration

All ten pinned-plugin cases passed: the interface exposed proposal and verification while retaining host admission, and its rejection and receipt-integrity controls matched the declared outcomes. Supplement S2.4 retains the lifecycle, unloading, and tamper details. This is interface-conformance evidence; the following live runs examine task execution on that interface.

8.8. RQ8: live task outcomes and separate host checks

The Final matrix contains all 1,260 planned executions. Among 360 planned benign runs, execution yield was 320/360 (88.89%) under the frozen strict rule and 331/360 (91.94%) under the post-hoc endpoint rule. Twenty-five runs had no scorable behavior verdict. Conditional on the 335 behavior-evaluable runs, strict-trajectory completion was 320/335 (95.52%; 95% exact interval 92.72–97.47%). Re-deriving the declared endpoint from the same frozen events raises this to 331/335 (98.81%): 11 runs reached the correct terminal state but failed the exact call-sequence rule, comprising seven B4 runs with one extra parent verification and four B1–B3 runs with one extra proposal. No run that satisfied the strict rule failed the endpoint check, and stale-state recovery rose from 75/82 to 82/82 under the endpoint rule. The increase measures the effect of relaxing the task criterion; independent corroboration would require separate evidence. Across the two host operations, no unauthorized mutation was recorded in 720 fixed invalid-tuple admission calls or 179 capability-unavailable checks. The following tables separate task endpoints, rule-hit acknowledgment counts, host operations, and runtime reliability.

Author-involved human annotation of the benign endpoint. One author (X.W.) and one non-author volunteer coded all 360 benign runs under the terminal-completion rubric. Original labels agreed on 359/360 runs (99.72%; three-category $\kappa = 0.974$, case-bootstrap 95% CI [0.911, 1.000]). Conditional on the 335 behavior-evaluable runs, both judged 331 terminally complete and each agreed with the strict rule on 324/335 (96.72%). The eleven human-positive/strict-negative cases reflect different criteria: extra calls do not violate the issued completion rubric. L.H. agreed with the strict-negative classification of the eleven discrepant cases; the frozen strict total remains 320/335, and the original completion labels are unchanged. T153, outside that comparison, was adjudicated separately. Supplement S2.8 retains binary

Table 14: Runtime-failure terminal classes crossed with endpoint evaluability.

Terminal class	Total	D1	G1	G2	Authority- evaluatable	Behavior- evaluatable
MODEL_API_FAILURE	44	35	0	9	27	0
INVALID_OUTPUT	37	37	0	0	37	0
HARNESS_FAILURE	11	4	4	3	0	3
TIMEOUT	1	0	0	1	0	0
Total	93	76	4	13	64	3

sensitivity, confusion matrices, unscored cases, procedure deviations, and clustering sensitivity; these are author-involved measurements.

Runtime failures totaled 93: 76/420 for D1, 4/420 for G1, and 13/420 for G2 (table 14). No `SETUP_FAILURE` or `TOOL_RUNTIME_FAILURE` occurred. Of the 93 failures, 64 retained complete authority verdicts; three harness failures remained behavior-evaluatable, all with task completion false. Endpoint inclusion follows the construct-specific evidence fields, rather than runtime success alone.

Table 15 retains configuration-specific evaluatable denominators. Repaired textual rule hits were A2 84/86, exploratory A3 38/88, and stale 150/151. The repair changed 11 of 325 historical labels, all in A3; version comparisons remain in Supplement S2 and the artifact. No unavailable-capability attempt was observed in evaluatable cells.

Blinded author annotation of textual acknowledgment. X.W.’s labels agreed with rule v2.1 on 286/325 outputs (88.00%; $\kappa = 0.644$, case-bootstrap 95% CI [0.543, 0.741]). A3 accounts for 32/39 disagreements and has no uniquely specified target field, supporting the construct-ambiguity limitation. All 39 disagreements remain without human adjudication; separate AI-assisted review proposals do not replace the frozen labels (Supplement S2). Supplement S2 gives per-scenario agreement, confusion matrices, additional coefficients, and resampling sensitivity.

All 25 unscored benign runs belong to recorded runtime-failure classes (table 16); the remaining 68 of the 93 runtime failures occur outside this unscored benign subset. If the 25 unknown behavior outcomes were assigned all failures or all successes, the strict proportion would lie between 320/360 and 345/360 (88.89–95.83%); the endpoint range would be 331/360 to 356/360 (91.94–98.89%). These are logical extreme-case bounds, not con-

Table 15: Agent-mediated outcomes (count/evaluable N). Strict verdicts are frozen; endpoint verdicts are post-hoc sensitivity results. Textual columns are deterministic rule hits, not validated semantic measures. A3 has no unique target and is exploratory.

Config.	Benign strict	Benign endpoint	Recovery strict	Recovery endpoint
D1	101/108	108/108	18/24	24/24
G1	114/119	115/119	29/30	30/30
G2	105/108	108/108	28/28	28/28

Config.	Stale rule hits	A2 rule hits	A3 exploratory hits
D1	32/33	26/26	3/28
G1	60/60	28/30	20/30
G2	58/58	30/30	15/30

Table 16: The 25 unscored planned benign runs, by recorded terminal class. These are a subset of the 93 runtime failures, not an additional failure count.

Terminal class	D1	G1	G2	Total
MODEL_API_FAILURE	9	0	8	17
HARNESS_FAILURE	3	1	3	7
TIMEOUT	0	0	1	1
Total	12	1	12	25

fidence intervals, recovered outcomes, or a missing-at-random assumption. Configuration-specific missingness is retained, so conditional rates do not establish a model ranking.

The 0/720 admission-call and 0/179 capability-unavailable results have distinct interpretations (table 9). Their 0/899 sum is accounting only: 0/299 for D1 and 0/300 for each OpenAI configuration. Table 17 gives the zero-violation counts by invoked mechanism family.

Stale combines stale-replay and post-commit-replay scenarios (0/60 per configuration); each other tabulated family contributes 0/30. Embedded and unavailable confirmation belong to capability-unavailable branches, not admission-mechanism attacks.

Task outcomes varied across the qualified configurations, while the predeclared host checks preserved authoritative state. Together they characterize task execution and host-enforced admission on the restricted interface. The fixed, model-independent challenge parameters delimit these observations to

Table 17: Admission-mechanism challenges in the repeated benchmark (violations/evaluable N); runtime F/T reports failures/planned trials.

Config.	Unauth.	Stale	Cross-rec.	Cross-fld.	Cand.	Evid.	Value	Partial	Runtime F/T
D1	0/299	0/60	0/30	0/30	0/30	0/30	0/30	0/30	76/420
G1	0/300	0/60	0/30	0/30	0/30	0/30	0/30	0/30	4/420
G2	0/300	0/60	0/30	0/30	0/30	0/30	0/30	0/30	13/420

integration conformance (section 9).

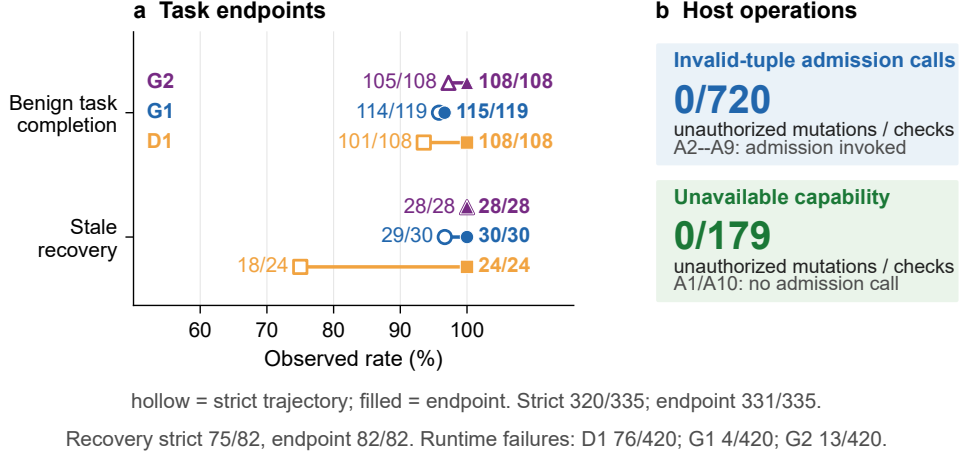


Figure 4: Task endpoints and authority outcome across three qualified configurations. (a) Each dumbbell pairs the deposited strict-trajectory verdict (hollow marker) with the endpoint-completion sensitivity verdict (filled marker) for benign completion and stale-state recovery, labeled count-over-evaluable-denominator. (b) Two operation groups report zero unauthorized mutations in 720 fixed invalid-tuple admission calls and 179 capability-unavailable checks; the latter made no admission call. Acknowledgment rule-hit counts are reported in table 15; their separate blinded author comparison does not turn them into externally validated semantics. The 93 runtime failures are reported separately.

9. Discussion and threats to validity

9.1. Design implications

The contribution is an inspectable relation among proposal origin, human value authority, and successor attribution. Table 18 turns C1–C2 into an implementation aid: map each class to persisted objects and an enforcement point, then retain a legal control and a failure probe. A check remains untested until that evidence exists. The worksheet is in `docs/admission-design-checklist.md`.

Representations may differ while satisfying the same obligations. D_B concerns the admitted change domain and complete successor; D_S concerns changed-field sources and exact unchanged-source retention. Full P6 tracing also follows authorization, candidate, and evidence bindings. The practical inspection question is whether an approval names the reviewed candidate or only its value. Complete audits can reconstruct sources without specifying

Table 18: Design checklist derived from C1–C2. Adapt the probes to the target implementation and retain persisted evidence for each outcome.

Class	Retain and check	Failure probe and expected evidence
D_C	Exact candidate, target, evidence and producer; bind the decision at review and recheck context inside admission.	Substitute an equal-valued candidate from another context: reject without authority writes. Reload the binding to verify candidate identity.
D_V	Proposal, human-authorized value and committed value with separate roles; preserve the candidate and check the latter two values at admission.	Authorize 101 from proposal 100: commit 101, retain 100 and its identity. Reconstruction must expose false machine attribution.
D_F	Expected and observed predecessor discriminator; compare within the transaction and protect it against races.	Supersede the predecessor before replay: reject without changing the new current state. Competing admissions from one predecessor must not both commit.
D_B	Admitted field domain, actual effects and commit grouping; validate the planned successor and commit it atomically.	Change two fields; interrupt between writes. Recovery must expose all or none of that admission, never an intermediate or fragmented successor.
D_S	One resolving source per successor field; validate changed-field sources and exact unchanged-source retention before commit and on trace.	Keep values fixed but remove, duplicate or replace a source. Reject invalid construction; diagnose incomplete trace after persisted corruption.

that review link; the checklist requires demonstrated relations, not a particular source-map representation. Its use in independent systems remains unvalidated.

9.2. Engineering trade-offs

Wider changes cost more on the frozen grid. The persistence-equivalent gap combines validation/planning and implementation-path differences, so it cannot price an individual check. Admission and trace require consistent canonical equality; the repair shares that relation, not a cached change set. Form-scoped tracing trades 12 database round trips for in-memory indexing and prefix traversal. The preferred access pattern remains workload-dependent.

9.3. Formal and concrete validity

Proposition 1 is conditional on the declared observation model. Its five class pairs, identity pair, and copy-forward control are constructions, not a general admission oracle. Alloy checks encoded scopes and selected mutations; the nine intended projections, twenty mapping mutants, catalogue, and SQLite schedules cover selected executions. The no-op and schema-uniqueness differences remain explicit in section 6.

Trace completeness checks persisted relations, not evidence truth, candidate accuracy, or human judgment. Independent oracles reduce shared-code risk but can share semantic defects, as the repaired equality bug showed. The adapter compares persisted digests and canonical locators without re-reading external evidence bytes. Detecting an out-of-band replacement that preserves metadata requires immutable/content-addressed storage or separate integrity verification.

Host authentication, faithful review presentation, serialization, hashing, policy, and adapter/transaction semantics form the trusted boundary (section 3). Review-channel manipulation, indirect instructions influencing approval, and adversarial confirmation inputs were not exercised. Binding preserves a recorded decision, not its quality. Other database engines need new conformance evidence. The commit-entry policy check rejects already-visible revocation but permits later-revoked in-flight transactions (section 3.5); it provides neither linearizable nor distributed revocation. Usability, authentication workflows, reviewer error, deletion, redaction, retention expiry, and tombstones remain untested; manual entry is outside the AI-derived claim.

9.4. Performance and repeated-agent validity

The empty-source comparator is a lower bound. Sequential trace timings may reflect caching, scheduling, checkpointing, and noise as well as access shape. Frozen cost results describe v8 and were not rerun after the equality repair.

The three qualified configurations from two provider families are a convenience population, not a basis for provider ranking. Endpoint names and settings do not identify immutable weights; provider load, decoding, and network conditions can vary. Pilot qualification did not ensure Final stability: D1 failures increased from 0/28 to 76/420. All planned runs remain included, with eligibility determined by table 14.

Terminal completion and strict-trajectory compliance are distinct constructs. The post-hoc endpoint criterion admits 11 strict failures; extra calls alone establish neither caution nor general recovery competence. Original benign labels came from one author and one non-author volunteer. The first author’s strict-trajectory review is a separate judgment, not an independent correction of terminal-completion labels.

The textual audit has one author coder and no human adjudication; the subsequent unblinded AI review supplies proposals only. Agreement is prevalence-sensitive and heterogeneous: A6 and B4 nearly coincide with the rule, whereas A3 contributes 32/39 disagreements. A3’s unspecified target limits the construct. Author involvement, knowledge of the study purpose, and residual textual clues constrain blinding; the audit supplies human–rule comparison rather than third-party validation. Frozen rule hits remain intact.

The 720 fixed invalid-tuple calls and 179 capability-unavailable branches measure the exercised host operations. They do not sample model-generated attacks, compromised hosts or identity providers, administrator access, or model access to admission. Zero violations imply neither a population failure bound nor general prompt-injection security; zero unavailable-tool attempts reveal no behavioral heterogeneity.

9.5. Reproducibility and evaluated-control boundary

The deposit supports inspection without new hosted calls; reproduction in another environment is a separate test. The related-work comparison covers disclosures through its stated cutoff. The executable controls establish a specific separation: both value-audit policies agree with candidate-bound admission on seven of eight single-history cases and reconstruct field sources,

yet leave the paired reviewed-candidate histories indistinguishable in authority-relevant state (section 8.5). Exact candidate binding preserves the review distinction left unspecified by these evaluated safeguards.

Richer review-context snapshots distinguish candidates when recorded proposal, producer, or evidence observations differ. Exact identity becomes material when distinct persisted candidates remain equivalent under those observations; it is not the only possible discriminator. This representational boundary leaves Proposition 1’s conditional D_V result intact: the controls test recoverability from particular audit representations, while the proposition tests omitted observation classes. Together they position an explicit correction-aware obligation, its conditional characterization, and its transactional realization. Independently developed event-sourcing/CQRS, temporal/bitemporal, and four-eyes systems were not evaluated.

10. Conclusion

Correction-aware authoritative-state admission connects proposal origin, human value authority, and complete successor lineage. Its practical test is the 90/100/101 history: can a later reader recover what the machine proposed, what the human authorized, and which sources the resulting record inherited? Under the declared observation model, five paired histories identify omissions that make the corresponding admission failures indistinguishable.

Bounded model checks, persisted-state projections, and transaction tests support the reference realization. Frozen cost measurements characterize v8, while repeated-agent results distinguish task criteria from host-enforced authority. The corrected implementation and its regression evidence remain separately versioned.

Developers applying the contract should preserve the exact candidate, record the human-authorized value separately, recheck policy and predecessor freshness inside admission, and atomically commit the complete successor with sources for every field. Changed fields receive their authorized transitions; unchanged fields retain exact prior sources. The checklist in table 18 and the worked example in Supplement S3 connect these obligations to implementation inspection and failure probes using existing transaction and audit facilities. Other database engines and operational reviewer workflows require further validation.

Declarations

Data and software availability

The study uses synthetic/public lifecycle inputs. The current manuscript, supplement, repaired code, formal models, observation witnesses, manual annotation, and derived analyses are archived at <https://github.com/lhh666-6/auto-dete/tree/r31-jss-2026-09-13/latest>. Its manifest covers the integrated package and an unchanged v8 manuscript snapshot. Frozen v8 experimental records retain their separate manifests at <https://github.com/lhh666-6/auto-dete/tree/r21-jss-2026-09-07-v8/r21-jss>; later repairs and review proposals do not replace those records. Reproduction instructions support inspection of the frozen evidence without paid model calls; rerunning hosted-agent calls requires provider credentials.

Ethics approval and consent to participate

The study used only synthetic and public lifecycle inputs. One author (X.W.) and one non-author volunteer independently coded benign completion outcomes; the first author (L.H.) adjudicated the inter-human disagreement and separately reviewed eleven human–rule disagreements against the strict-trajectory criterion. X.W. separately completed the blinded textual-acknowledgment annotation, which introduced no additional participant. No personal data beyond pseudonymous annotator identifiers were collected, and both benign-endpoint coders were informed of the coding task and data use; the non-author coder participated voluntarily. No animal subjects were involved. Annotation evaluated generated outputs rather than annotator characteristics. The authors assessed this activity as not requiring ethics approval.

CRedit authorship contribution statement

Liang Hanghao: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Validation, Visualization, Project administration, Writing–original draft, and Writing–review & editing. L.H. led the core research contributions, designed most experiments and the blinded annotation protocols, and wrote the original manuscript.

Xuan Wentao: Methodology, Formal analysis, Investigation, Data curation, Validation, Visualization, and Writing–review & editing. X.W. contributed to the design and statistical analysis of the non-annotation experiments supporting C3 and C4, participated in blinded annotation, and refined the manuscript

language and flowcharts. X.W. completed the 325 textual-acknowledgment labels and rationales; the annotation protocols were designed by L.H.

Peng Peng: Supervision and Writing-review & editing. P.P. provided research guidance, recommended relevant literature, and supplied critical comments on the manuscript.

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Declaration of competing interest

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During manuscript preparation and revision, DeepSeek assisted with a first revision of code, analyses, figures, and prose; OpenAI Codex assisted with literature organization, evidence-linked drafting, deterministic table and analysis code, observation-witness checking, language editing, and integration of the completed manual annotation. Claude assisted with the executable-example runner and its initial integration; Codex subsequently reviewed and repaired that integration. The repeated benchmark exercised two requested OpenAI configurations through Codex CLI and one requested DeepSeek configuration through its API; model identifiers, prompts, raw events, tool calls, host actions, runtime classes, and usage metadata are retained. X.W. completed the 325 manual labels and rationales before the human-rule analysis; no AI system assigned or altered those labels. Codex subsequently reviewed the 39 human-rule disagreements without blinding and recorded proposed classifications and quotations in a separate artifact; these proposals are not human adjudications and do not replace any original label or reported agreement statistic. For the September 2026 figure-script revision, the assistant was OpenAI Codex using model `gpt-6-astra` (developed by OpenAI); rendering used deterministic Matplotlib code rather than generated image pixels. Precise assistant versions for earlier preparation stages were not archived. Research decisions, source verification, interpretation, and final responsibility remain with the authors. An AI-generated raster was used only for internal ideation and is not included. Figures 1 and 2 were programmatically redrawn with Codex assistance from the authors' approved workflow, verified contract, and contribution definitions. Figures 3 and 4 are

regenerated from the same frozen data with a legibility revision (enlarged labels, dense heatmap values moved to the complete supplement table, and the strict/endpoint task verdicts separated); frozen experimental records were retained, while the added endpoint sensitivity analysis, repaired rule-hit counts, and blinded author annotation are separately versioned.

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